

BE 504: Biomaterials

Course Code	BE 504
Course Name	Biomaterials
L-T-P-C	3-0-2-4
Prerequisites	IC136, IC241, or consent of faculty
Intended for	B. Tech M.Tech Integrated Dual Degree Bioengineering students
Distribution	Core for IDD BE, Elective for other B.Tech students
Approval	44th BoA (Nov 2021)
Revised	49th BoA (Mar 2023) - revised

Course Contents

- **Introduction to Biomaterials**

Introduction to biomaterials and its history, Properties of Biomaterials- physico-chemical, mechanical, biocompatibility and biodegradability, Surface properties of biomaterials, Biological responses and Cell-Biomaterial interaction. [8 Lectures]

- **Classes of Biomaterials**

Polymeric materials and blends, Biopolymers and hydrogels, Metal based biomaterials, Ceramics and bioglasses; Adhesive and sealants, Elastomers. [8 Lectures]

- **Biomaterials Applications**

Applications of biomaterials in cardiology, nephrology, ophthalmology, dentistry and orthopaedics; Wound healing and dressing materials, skin substitutes and sutures, Applications of Biomaterials in Functional Tissue Engineering and drug delivery systems. [20 Lectures]

- **Device development, Standards and regulatory compliance**

Biomaterial device development and Regulation, Voluntary consensus standards, Commercialization, corporate considerations, Ethical issues, Clinical trials, Entrepreneurship and post market considerations in biomaterials. [6 Lectures]

Laboratory Modules

Lab component of the course will include synthesis/fabrication, characterisation, biocompatibility testing and application of biomaterials. The experiments include.

- Preparation of polymeric nanosphere and characterization
- Drug/Protein loading and release study with polymeric nanospheres
- Preparation and characterization of hydrogels/nanosponges
- Preparation and characterization of electrospun nanofibrous mats
- Biocompatibility Testing
- Preparation of Bioceramics and its characterization

Text Books

1. Editor(s): Buddy D. Ratner et al, **Principles of Biomedical Instrumentation** in Biomaterials Science, 3rd edition, Academic Press.
2. Bikramjit Basu, **Biomaterials Science and Tissue Engineering: Principles and Methods**, Cambridge University Press, 2017.

References

1. Bikramjit Basu, **Biomaterials for Musculoskeletal regeneration: Concepts**, Springer Nature, 2017.
2. **Advanced Biomaterials: Fundamentals, Processing and Applications**, John Wiley & Sons, 2009.
3. Fredrick H. Silver and David L. Christiansen, **Biomaterials Science and Biocompatibility**, Springer.
4. Related journal articles.